SPE-227531-MS

Using Colored Inversion in the Geosteering and Well Placement: Bridging the Gap Between the Operations Geoscientists and Asset Geoscientists

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

Development of unconventional reservoirs requires optimum placement and geosteering of lateral wells through the target zone. In many instances, due to variable lithology and reservoir properties, well placement encounters many challenges thus necessitating more detailed studies. In many occasions, particularly during the appraisal phase of the reservoir, the lateral wells are preceded by the vertical pilot wells in order to accurately identify the reservoir zones and successfully guide the lateral well through the target. However, in the absence of pilot wells, the task to identify a suitable landing point and subsequent geosteering of the well becomes a challenging task. In order to alleviate the challenges so as to run a more successful well operation to optimize the cost and production, we propose a fast-track seismic-based approach to well design and planning. Our approach is based on the colored inversion attribute, which can be utilized to accurately identify the location and thickness of the subsurface target zones. The attribute is used to geosteer the well through the target zone, thus maximizing the possibility to effectively encounter the sweet spots along the lateral. In addition, we have used the colored inversion attribute, augmented with a well-driven low frequency model to accurately predict the total porosity of the reservoir.

Introduction

Extraction of resources from unconventional reservoirs requires a detailed plan for the drilling program. The drilling program could face many challenges particularly when the target reservoir is thin and tight. In such cases, purely seismic-driven well planning suffers from uncertainties due to the lack of seismic resolution, resulting in the tuning effect. A thin and seismically-tuned reservoir results in greater uncertainties in terms of its thickness when interpreted from conventional seismic data. Moreover, it is well-known that seismic resolution is a function of frequency bandwidth of the seismic data. Conventional seismic data lacks the low frequency components, which reduces the bandwidth, making the interpretation less accurate. In addition, varying lithology along the length of the lateral adds to the complexity of the seismic data, both in amplitude and travel time. Furthermore, unconventional reservoirs, particularly the deeper ones, are known for their low porosity. The high pressure and temperature environment in the deeper levels diagenetically alter the rock composition and fabric, which in many instances result in quartz overgrowth and/ or illite coating of the grains, resulting in significant reduction in porosity.

It is known that as the seismic wave propagates deep into the subsurface, the higher frequency components get attenuated faster. As our target reservoir is deep, the combined effect of lack of low and high frequency components in the seismic data result in low seismic resolution. In addition, our reservoir is expected to have inherent lithological heterogeneity, making the interpretation from conventional seismic data challenging.

Colored inversion (Lancaster and Whitcombe, 2000) is a proven workflow commonly used to enhance interpretation accuracy, particularly when the seismic data is low in resolution and the reservoir is complex. Like other seismic inversion approaches, the colored inversion workflow converts a reflection event, an interface property in the seismic data, to its corresponding layer property. Such a mapping from the interface property to the layer property enhances seismic interpretability by increasing the low frequency components in the colored inversion attribute. Enhanced accuracy in the seismic interpretation adds a lot of value in subsequent well planning and design.

Inversion workflows convert the reflectivity seismic data to the corresponding impedance domain. Since colored inversion is an attribute derived from the seismic data with inadequate low frequency information, it doesn't contain information about the absolute impedance variations within the layer. For this reason, colored inversion is also known as relative impedance inversion, as the attribute shows only the relative variations in the layer impedance. Even in the absence of absolute impedance, the workflow is highly useful because it is quick and easy to implement and the output product is easy to interpret with higher accuracy compared to the interpretation of conventional seismic data.

Theory

Colored inversion is a fast-track seismic inversion method that computes an operator (also known as the colored inversion operator), which, when convolved with the seismic data, alters the frequency spectrum of the seismic data to approximate the frequency spectrum of the acoustic impedance logs at the well locations. In practice, the impedance logs and the seismic data used for the calculation of the colored inversion operator are cropped around the target zone that include a sufficient buffer zone above and below the target interval. The colored inversion workflow is fast-track because it doesn't involve performing a full model based inversion which needs a low frequency model to be built and wavelets to be estimated. Colored inversion is commonly used to enhance the vertical resolution and interpretability of seismic data. This is achieved as the colored inversion operator enhances the bandwidth by incorporating frequency components of the impedance logs, particularly in the low frequency end of the spectrum.

The premise is that the seismic data is inherently band-limited version of the Earth's true reflectivity. The band-limitation is caused by the presence of the wavelet that commonly has a bell-shaped spectrum. The wavelet in the data carries the combined effects of the source signature, attenuation through the medium, and recording limitations. The colored inversion workflow tries to replace the band-limited reflectivity signature in the data with the corresponding relative impedance within the layer by shaping the spectral signature of the seismic data to match that of the impedance log. The spectral shaping is carried out within a reasonable signal bandwidth. The workflow involves the following steps:

1. Ensure that the seismic reflectivity data is zero-phase. If not, the data needs to be phase-rotated to correct for the spurious phase.
2. Extract the seismic amplitude spectrum from a representative time window encompassing the target interval.
3. Estimate the target impedance spectrum from the well impedance logs.
4. Design a spectral shaping operator (or filter) that transforms the seismic spectrum into the target impedance spectrum.
5. Introduce -90° phase rotation to the colored inversion operator to mimic the phase spectrum of the impedance logs.

6. Apply the operator to the seismic data trace-by-trace, which will result in a spectrally altered dataset that mimics the impedance log spectrum. This operation can be achieved either by frequency domain multiplication or time domain convolution.

This method enhances, within the limits of the signal bandwidth, both the low and high-frequency ends of the spectrum, resulting in enhanced resolution and better event continuity, which in turn, improves the interpretability of the seismic data.

Workflow

It is essential to evaluate the quality of the seismic data before implementing the colored inversion workflow. The quality of the data was assessed on the basis of spectral bandwidth and phase spectrum around the zone of interest. In addition to the wedge modeling and borehole synthetics, the data quality was evaluated by carefully analyzing the accompanying VSP corridor stacks. The wedge model and the well synthetics revealed that the seismic amplitude at the reservoir top is a composite of two distinct interfaces- the sealing tight formation and the low porosity reservoir. The tuning effect between the two interfaces have resulted in a deformed seismic amplitude. However, as the overburden and the sealing rock were field wide uniform, we decided to proceed with the recorded seismic amplitude without applying any de-tuning workflow. In addition, to further ensure the reliability of the data, a seismic-to- well tie was conducted. This process allowed us to evaluate the quality of the seismic data and quantify any discrepancies in amplitude and phase responses. The correlation coefficient, which measures the match between synthetic and actual seismic data, was found to be approximately 0.7 within the interval of interest. The 180° phase-rotated wavelet was estimated from the data for the well synthetic generation. As evident from Figure 1, the phase in the data is very close to zero, which suggests that the data meets the zero-phase requirement criterion for colored inversion. This reasonable match indicated a reliable signal-to-noise ratio, with the primary energy dominating the data. Figure 2 shows the zone of interest in the log domain, along with the well synthetic seismic data, surface seismic data, and the VSP corridor stack. A good match is observed between the different datasets, reinforcing the confidence in the dataset's suitability for colored inversion and its potential to yield meaningful insights into the subsurface characteristics.

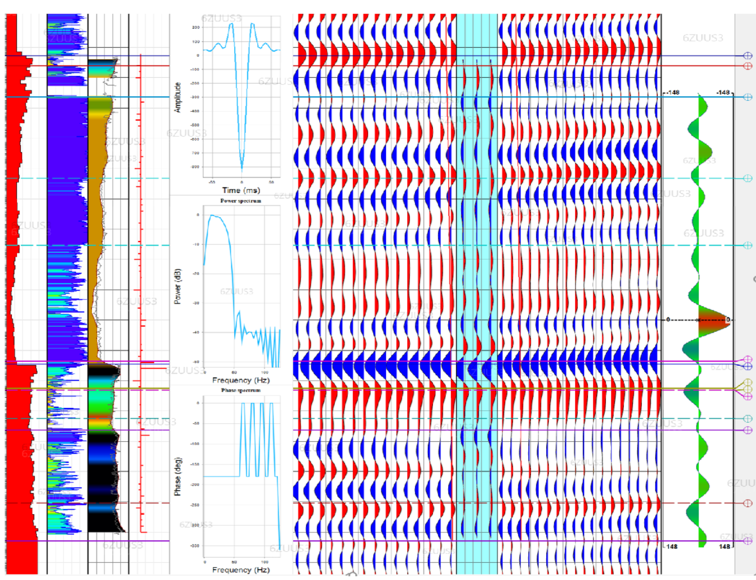


Figure 1—Statistically extracted wavelet used to construct a primaries-only synthetic seismic for well tie. A reasonable match with the surface seismic data is evident from the figure. The correlation coefficient is 0.7. Also, the data appear to be very close to zero phase.

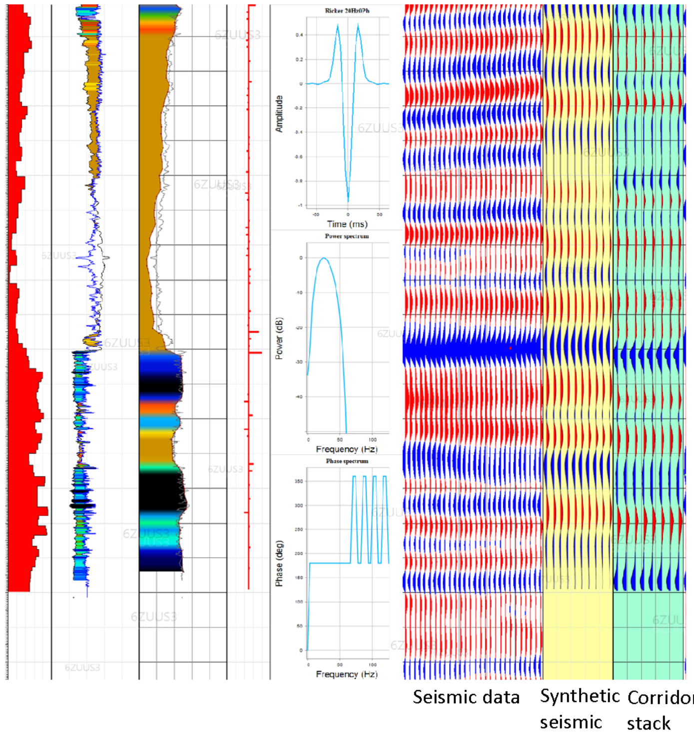


Figure 2—The target interval, surface seismic data, the synthetic seismic, and the primaries-only VSP corridor stack. A reasonable correlation is observed, which confirms the reliability of the seismic data for inversion.

The following workflow was adopted to design the colored inversion operator and apply it to obtain the colored inversion output.

1. QC of the sonic and density logs.
2. Log conditioning to remove outliers and bad-hole data.
3. Transform the impedance log to the frequency domain.
4. Select the seismic data limits.
5. Compute an average amplitude spectrum for the data around the zone of interest.
6. Define the seismic bandwidth for a reasonable signal-to-noise ratio.
7. Calculate the colored inversion operator that transforms the seismic average spectrum to log impedance average spectrum within the defined bandwidth.
8. Apply a −90° phase rotation to the colored inversion operator.
9. Convolve the colored inversion operator with the seismic data to obtain the final relative impedance output.

Figure 3 – shows a schematic diagram explaining the workflow to calculate the colored inversion operator.

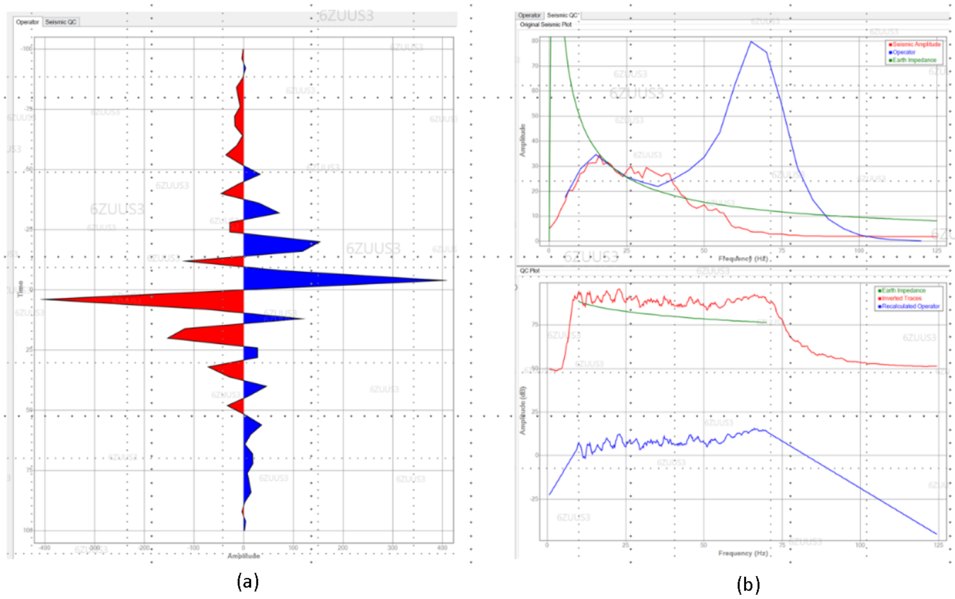
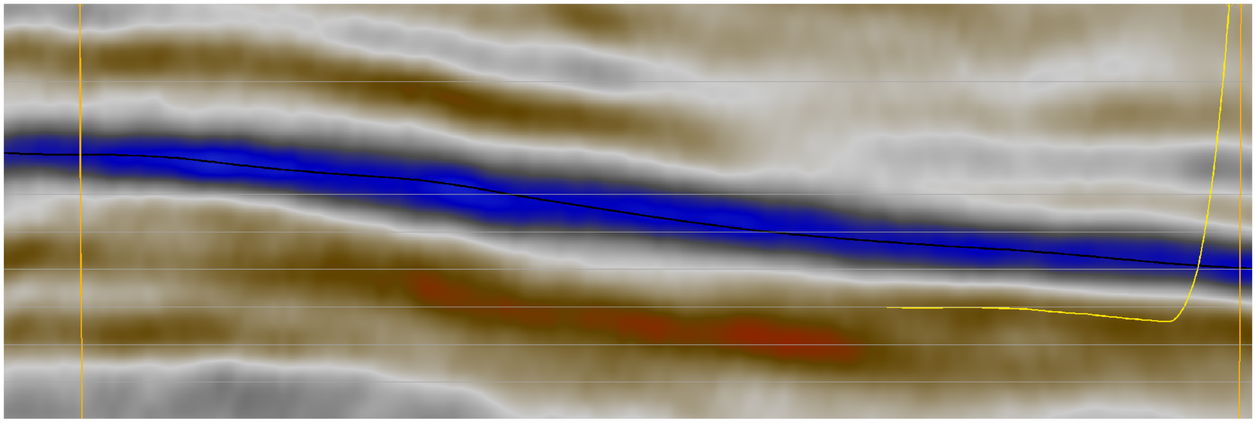
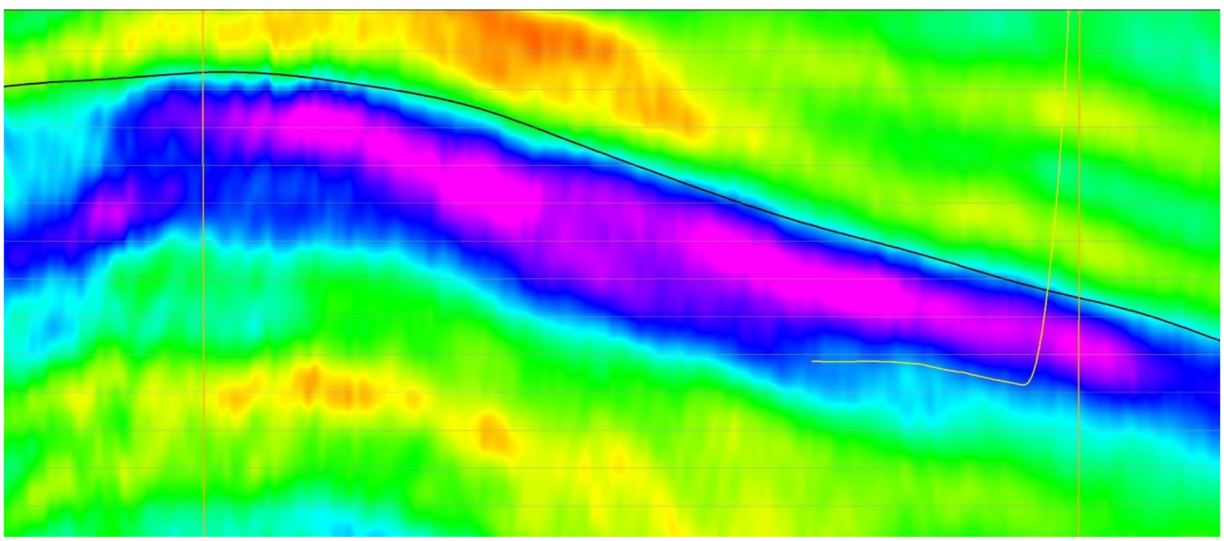


Figure 3—Colored Inversion operator in (a) time domain and (b) in frequency domain derived by calculating a shaping operator that transformed the seismic spectrum to match with the log impedance spectrum. The operator is subsequently convolved with the seismic data to produce the bandlimited acoustic impedance.

The colored inversion volume was subsequently used in the seismic-guided geosteering of the lateral wells. Figure 4(a) shows the seismic reflectivity data and the top of the reservoir grid superimposed. Figure 4(b) shows the colored inversion volume derived from the seismic reflectivity data. It is observed that the colored inversion volume has transformed the reflectivity data (an interface property) into the relative impedance domain (a layer property), as a result of which the layers below the reservoir top are clearly seen and the event continuity is better preserved. The colored inversion volume helps in guiding the well trajectory more effectively and accurately.



(a)



(b)

Figure 4—(a) Seismic reflectivity data used to generate the colored inversion volume. (b) A zoomed-in view of the colored inversion volume. The layer structure in the zone of interest is well visible, which significantly helped in the geosteering of the lateral well. The well paths are shown in the yellow lines on the seismic and colored inversion sections.

As mentioned earlier, the colored inversion volume yields relative impedance attribute, which indicates a relative variation in acoustic impedance when compared with the adjacent rock types. The relative variation shows whether or not a particular rock unit is softer or stiffer than the adjacent rock units. However, it is always desirable to obtain the impedance volume in the absolute domain so that reservoir properties, such as porosity, can be predicted with a much better accuracy. An absolute impedance volume can be obtained from a colored inversion volume by adding the missing low frequency components which can be obtained from the well logs. Hence, as a second step, a low frequency model was built using the well impedance logs, which was subsequently added to the colored inversion volume generated in the earlier step. The resulting absolute impedance volume provides a faster way to obtain an output equivalent to the output of the model-based inversion. The absolute impedance volume was then used to predict the total porosity distribution within the zone of interest. In order to predict the porosity from the impedance, a rock physics transform equation is needed. The equation is obtained by cross-plotting the impedance logs and the associated total porosity logs for the available wells and computing the equation that fits a line to the cluster of the points in a least-squares sense. Figure 5 shows the cross-plot between the impedance and the associated total porosities in the well log domain. The best-fit line is also shown on the cross-plot.

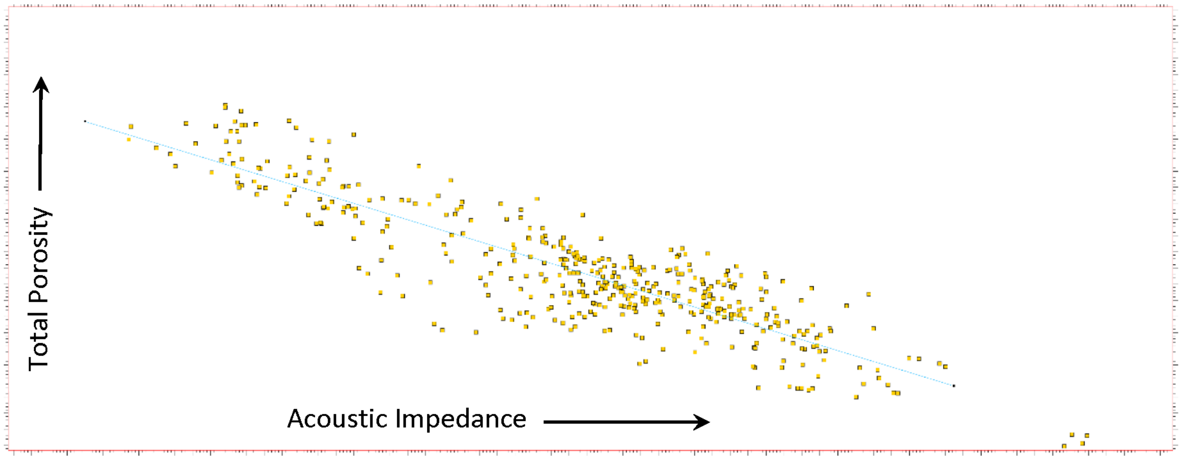


Figure 5—Cross-plot between the total porosity and acoustic impedance in the well log domain. The best-fit linear regression line is shown superimpsed on the cross-plot. The linear equation for the regression line is used to transform the inverted acoustic impedance in the absolute domain to total porosity.

The well-derived linear regression equation was subsequently applied to the absolute impedance volume to obtain the predicted porosity volume over the entire zone of interest. Figure 6 shows the absolute impedance section obtained by adding the well-guided low frequency impedance model to the relative impedance volume generated by the colored inversion. For the purpose of comparison, the acoustic impedance log is overlain on the absolute impedance section. The log and the section have the same color scale. It is clearly seen that the inverted absolute impedance matches reasonably well with the impedance computed in the log domain. Figure 7 shows the predicted total porosity section with the total porosity logs overlain. The predicted porosity is computed by applying the transform equation to the absolute impedance volume computed earlier. It is seen that, when compared with the log porosity, the total porosity is predicted with a reasonable degree of accuracy.

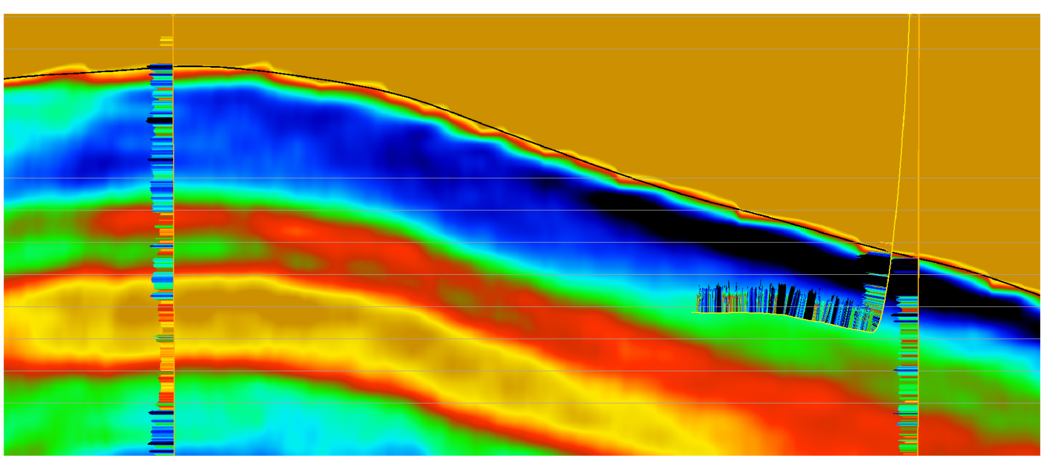


Figure 6—The predicted total porosity section obtained from the absolute impedance derived from the colored inversion attribute. The total porosity logs are superimposed. The logs and the porosity section have the same color scale. It is clearly evident that the predicted total porosity matches with the well log porosity with reasonable accuracy.

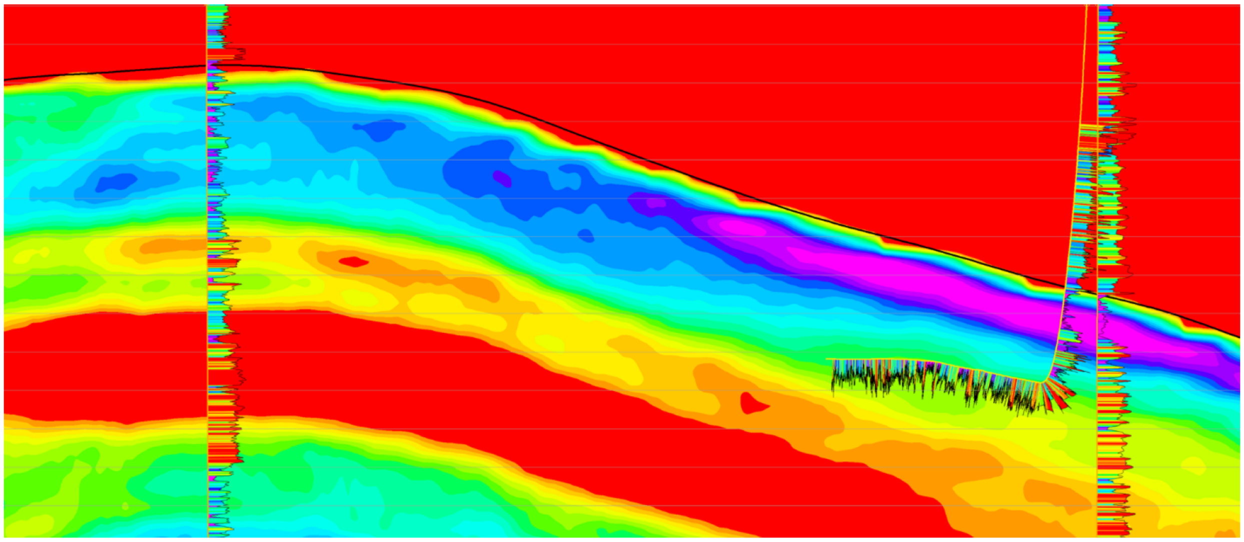


Figure 7—The predicted total porosity section obtained from the absolute impedance derived from the colored inversion attribute. The total porosity logs are superimposed. The logs and the porosity section have the same color scale. The accuracy of prediction is clearly evident from the match seen with the log porosities.

Discussion and Conclusions

Initially, the landing point and the well trajectory for the well under study were determined using the gamma ray and the total porosity logs at the pilot well location. However, subsequent analysis with the colored inversion attributes revealed that the location and trajectory were sub-optimal and needed to be re-visited. Accordingly, after re-evaluation, it was decided to place the landing point deeper than the earlier position. Subsequent mud-gas response indicated that the re-positioning of the landing point and re-designing of the well trajectory resulted in the well being efficiently steered within the sweet-spot interval, which enhanced the chance of success.

Furthermore, we observed a good correlation between the total porosity and the normalized rate-of-penetration (ROP), which helped us to qualitatively assess the drilling performance. The implications of accurate prediction of ROP is profound in terms of rig time, bit runs, tripping time and the overall well cost. By integrating inversion results with real-time data, we facilitated collaboration among geomechanics engineers, drillers, operations geologists, and geoscientists. This collaboration enabled us to make informed decisions, such as altering the well landing point and the trajectory to softer beds, thereby accelerating ROP and improving the drilling performance while reducing non-productive time (NPT).

The colored inversion workflow has proven to be an effective tool for well placement and geosteering of the lateral wells. The relative acoustic impedance inversion attribute effectively captures the layer zonation, which are otherwise not clearly evident in the reflectivity seismic data. In the study area, we not only identified three distinct zones - tight, moderately porous, and relatively more porous zones - but also identified stratigraphic pinch-outs that were not previously seen in the seismic data. Apart from the benefits in terms of better seismic interpretation, we also found that the colored inversion can be effectively used to predict the total porosity within the area of interest, which in turn can help in predicting the normalized ROP for better well planning and overall cost effectiveness.

The efficiency of the colored inversion workflow depends significantly on the quality of the input seismic data and the impedance logs. To start with, a good well tie is very important to obtain a reliable colored inversion and the derived attributes. Signal-to-noise ratio, interference from the multiples, and limited bandwidth resulting in tuning of the data are some of the parameters that need to be carefully studied before implementing the workflow.

The quality of the well log porosity data plays an important role in the prediction accuracy of the total porosity from the impedance attribute. Different best-fit trend lines, such as the power function and exponential function, may also be tried in order to determine the most accurate transform function between absolute impedance and total porosity.

With regard to future work, the colored inversion workflow can be designed to incorporate pre-stack data. The colored inversion attributes obtained in the pre-stack domain may be further used in the overall context of the Extended Elastic Impedance (EEI) workflow to obtain fast-track predictions of other reservoir properties such as the lithology types and water saturation.

Reference

S. Lancaster, and Whitcombe, D. (2000). Fast-track ‘coloured’ inversion. *SEG Technical Program Expanded Abstract*, 2000.